Yilin Wang (27Fall)

Basic Information

Phone/WeChat: 13148035071 Email: yilinw376@gmail.com, 13148035071xjtu@stu.xjtu.edu.cn

Github: [Homepage] Research Interests: Knowledge Conflicts in LLM, RL4LLM Reasoning, Interpretability

Education

2023.09-Present (Now third-year student) **Xi'an Jiaotong University**-School of Computer Science (Honors Science Program) Xi' an China

- **Academic performance:** grade point ranking **8/45** (first year), **12/40** (second year); credit-weighted score: **91.83**; GPA: **3.93/4.3**.
- **Coursework:** Computer Programming 100, Discrete Mathematics 97, Probability Statistics and Random Processes 100, Fundamentals of Optimization Methods 98, Linear Algebra and Analytical Geometry 98, Operating Systems 95, Mathematical Physics Equations 97, etc.
- English: CET-4: 622, CET-6: 536

Research

Co-lead the undergraduate research group of Professor Minan Luo (LUD) at Xi'an Jiaotong University with research outputs:

2024.05-2025.02 **Continuously Steering LLMs Sensitivity to Contextual Knowledge with Proxy Models** **EMNLP 2025 Oral**

- TL;DR: We propose the CSKS framework to address LLMs' difficulty in appropriately adjusting their belief in parametric knowledge when it conflicts with retrieved context in RAG. CSKS leverages the output distribution difference between two small models to guide the large model's predictions. Experiments show that CSKS significantly outperforms mainstream baselines on knowledge conflict handling. Code is open-sourced [Link].
- Contribution: **First author,** independently responsible for project design, coding implementation, and paper writing.

2025.03-2025.12 **Why Knowing Both Hops Is Not Enough: Understanding Two-Hop Generalization in Language Models** **Arr March Under Review**

- TL;DR: We develop a symbolic intervention framework and applied mechanistic interpretability to show that LLMs fail out-of-distribution in two-hop reasoning due to high-level functional mismatch, and proposed a recurrent training scheme that aligns layer-wise representations to achieve robust multi-hop reasoning across distributions.
- Contribution: Served as both **co-author** and **mentor.** I proposed the methodology, built the code framework, and led paper writing as co-author, while mentoring a collaborator to complete the remaining tasks.

Advised by Prof. **Wen Muning** at SAI, Shanghai Jiao Tong University; research on designing RL algorithms to enhance LLM reasoning.

2026.01-Present **Think About LLM as Pi**

- TL;DR: We model the forward inference process of LLMs as an MDP and aim to exploit internal model signals for algorithmic optimization.
- Contribution: **Research assistant; proposed the project methodology.**

Projects

2025.08-2026.02 Completed Stanford CS336 and most course assignments independently.

- Introduction: CS336 covers the full LLM stack from fundamentals to cutting-edge techniques, including model architectures, pre-training, post-training, and inference optimization.
- Contribution: Independently implemented most assignments with heavy, engineering-focused code, gaining solid practical and theoretical understanding of LLMs; code is open-sourced [Link].

2026.01-Present To learn the popular RL framework verl, I started developing a lightweight verl-style framework nanoverl

- Introduction: Scaling RL heavily relies on frameworks like verl. To understand framework design and more easily customize code for research needs, I began developing nanoverl.
- Contribution: Independently developing the project; a complete framework is now in place, supporting FSDP-based distributed RL training. The code is open-sourced [Link].

Honors

- First Prize (National Level) and Innovation & Entrepreneurship Award, 18th National College Student Information Security Competition
- Huawei Scholarship for Undergraduate Students
- First Prize, National College Student Artificial Intelligence Security Competition
- Nomination Award, Cybersecurity Innovation and Entrepreneurship Competition
- Third-class Scholarship (University-level)

FYI

- In my spare time, I enjoy playing badminton and volunteering, and served as Head of the Empty-Nest Outreach Department in the XJTU Yangfan Student Society during my first two years.